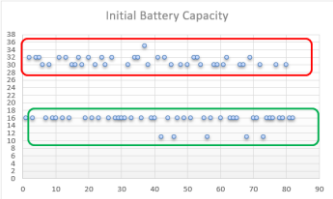
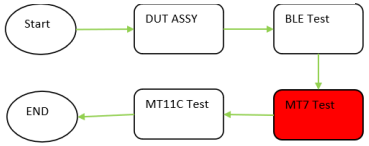
Beta Charger Initial SOC Value Abnormal(User Mode)-2.7%

1. Background

1. Read SOC Value (Exit MT)
1. What: The Beta Charger initial value in user mode of 47.56% DUT is 30% or above at MT7, but the high limit at MT11C is 30% and vendor shipped the battery with 25% SOC.
2. When:
- 2025/12/2 – 39 pcs units read value abnormal
3. Who: Beta Charger MT7
4. How many: 39/82, read abnormal rate 47.56%.
5. Why: DUT FW update initial SOC value is abnormal, the value is high than actual value.

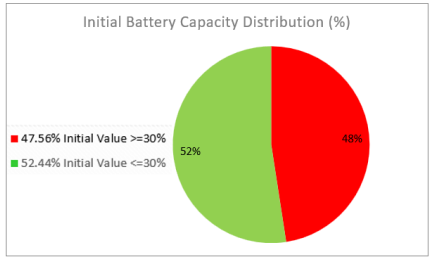




2. Current Situation

The Issue is detected in Engineering Build No Yield Impacted:


```
{
  "pnum": "4082",
  "reference": "BATTERY_PERCENTAGE",
  "time": "2025-11-28T17:23:33+08:00",
  "status": "pass",
  "unit": "%",
  "min": "",
  "max": "",
  "actual": "32"
},
```

Input 82 pcs in MT7BLE station, 39 pcs with this initial SOC value abnormal issue

3. Root Cause Analysis

1. Double checked the SOC value that read at MT11C, the value is same as initial value at MT7.
2. Since the SOC value is updated by FW based on the battery voltage level, compared the SOC value and battery level, and found that the initial SOC value is abnormal, the SOC value is higher even the read battery level lower for the abnormal units, see table as below for more details.
3. Feedbacked this issue to RD team, they investigated and found that some units may need more hard delay time for waiting system initialization after exit MT mode for Beta product, the hard delay times needed is 1.7s~2.3s, but test software set hard delay times is 2s based on Hermes experiences.

Date_Time	codentify_code	Battery_current	Battery_Capacity	Battery_Level
2025-11-28 13:17:50,376	T4E7 AWD 61V 1XJW	0 400 0	255 16 16	3584 3624 3564
2025-11-28 17:23:42,175	T52T 9VA MGW FF1Z	0 350 0	255 32 32	3584 3584 3584

Conclusion:
Beta product need more hard delay time for system initialization success.

1. Increased the test software hard delay times to 6s from 2s, the initial SOC value become normal

4. Goals and Objective

5. Proposed Improvement

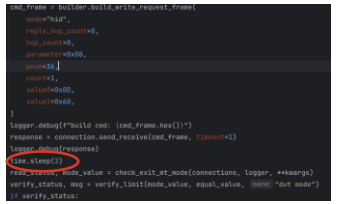
1. Increase the hard delay times to 6s from 2s as RD team proposal.
- 2025/12/4 - Input 148 pcs, no more this initial SOC abnormal units be detected.

2. Since increasing the hard delay times to 6s from 2s will extend 4s total test time. It will affect test UPH seriously, so try reduce hard delay time to 4s from 6s.
- 2025/12/6 - Input 555 pcs, no more this initial SOC abnormal units be detected also.

3. Based on above 2 step validation and confirmed with team, propose increase the hard delay times after send exit MT commands from 2s to 4s
- 2025/12/7~ 2025/12/14 Input around 1500 pcs, no more this failure captured.

6. Improvement Plan

Details test software updates as below.



7. Monitoring

1. To monitor the result of each build.